

PRATYUSH PRAKHAR

Delhi, India | +91-7042236696 | pratyushprakh@gmail.com

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PROFESSIONAL SUMMARY

Cybersecurity Engineer with 4+ years across AppSec, cloud security, and GenAI security. Built 3+ years of depth across the AWS cloud security stack spanning AWS direct and AWS-focused consulting, securing 15+ high-risk services, remediating 150+ critical vulnerabilities, and reducing attack surfaces up to 30% via STRIDE threat modeling and zero-trust principles. Early contributor to Amazon Bedrock GenAI security, including OWASP LLM Top 10 guardrails, prompt injection defense, and adversarial ML controls. OSCP-track. Applying security and AI rigor to financial research workflows in private and regulated markets.

TECHNICAL SKILLS

Application Security: SAST/DAST, API Security, STRIDE/PASTA Threat Modeling, OWASP Top 10, OWASP LLM Top 10, Secure Code Review, Vulnerability Management, CVSS, CVE Management, SSDLC, AI Red Teaming, Zero-Trust Architecture

Cloud & Infrastructure: AWS (IAM, S3, EC2, GuardDuty, Security Hub, CloudTrail, KMS, Amazon Inspector, AWS Config), CI/CD Security, SIEM/SOAR Implementation, Secrets/Token Management, PCI-DSS, NIST CSF, ISO 27001, SOC 2

AI/GenAI Security: LLM Security, Prompt Injection Defense, Adversarial ML Defense, Memory Poisoning Defense, Tool Misuse Controls, RAG Security, Output Validation/Sanitization, Model Risk Assessment, AI Supply Chain Security, NIST AI RMF, AWS Bedrock Security, LangChain/LangGraph, CrewAI, PydanticAI, Pi Agent, Claude Code, Codex

Offensive Security: Penetration Testing (OSCP methodology), API Penetration Testing, Burp Suite, Metasploit, Nmap, Nessus, Wireshark, SQLMap, Hydra, MITRE ATT&CK Mapping

Languages & Databases: Python, Golang, TypeScript, JavaScript, Bash/Shell, SQL, HTML/CSS, MySQL, DynamoDB

Regulated Domain Exposure: Financial Data Governance, Audit Reports, Private/Unlisted Markets, RBI Cybersecurity Framework, DPDP Act, EU AI Act

CERTIFICATIONS

- AI Security Fundamentals | Microsoft Learn
- Security for AI Essentials | IBM SkillsBuild
- AI Fluency: Framework & Foundations | Anthropic Academy
- OWASP Top 10, OWASP LLM Top 10 | PortSwigger
- Web Pentesting, Burp Suite, Metasploit | TryHackMe
- OSCP Preparation | Offensive Security methodology | Expected Dec 2026

PROFESSIONAL EXPERIENCE

FinTech Research Analyst | Bridging Gap Investment Services, Mumbai

October 2025 - Present

- Building AI-assisted research pipelines to ingest and structure company financials, audit reports, regulatory filings, and management signals, converting fragmented manual tasks into repeatable internal intelligence workflows that streamline deal research and ongoing market monitoring.
- Designed data governance and security controls for private and unlisted market data pipelines, applying access controls, encryption, and audit-ready architecture to ensure data integrity, regulatory compliance, and operational traceability across a regulated private-market research environment.
- Produce investment memos, market landscape reports, and client-facing research covering pre-IPO, pre-placement, and primary market instruments for institutional investors and broker clients.

Sr. Security Analyst | Solgenie Technologies Inc., USA

January 2024 - December 2024

- Continued AWS-focused security work across enterprise clients in financial services and SaaS, identifying critical vulnerabilities, building risk-indexed remediation plans, and reducing cross-account AWS attack surface by 20% through targeted API security and IAM controls.
- Delivered end-to-end AppSec consulting, vulnerability assessments, secure code reviews, and OWASP Top 10 compliance across API and microservice architectures, reducing security assessment turnaround by 40% and saving \$10K/month in operational costs.
- Automated SAST triage and incident response workflows using Generative AI, improving assessment accuracy by ~10% and enabling faster remediation cycles across cloud-native environments.
- Standardised penetration testing practices across 4+ client engagements using Burp Suite, Metasploit, and Wireshark, establishing repeatable security baselines and elevating client security posture.

Application Security Engineer | Amazon Web Services, New York, USA

July 2021 - December 2023

- Owned security design review authority for high-risk AWS services including CodeSuite, IAM Console, and SDKs, enforcing global security best practices across 15+ services and driving multi-framework compliance (NIST CSF, ISO 27001, PCI-DSS, SOC 2) through systematic vulnerability triage and architectural control design.
- Led AppSec initiatives for Amazon Bedrock's Generative AI platform across the full preview-to-GA arc, designing threat models and security controls (including memory poisoning defenses, tool misuse controls, and prompt injection guardrails) for AI infrastructure used

by enterprise customers in regulated industries globally. Implemented adversarial ML defenses aligned to OWASP LLM Top 10, reducing open high-risk findings by 25% across four iterative assessment cycles.

- Performed whitebox, greybox, and blackbox security assessments to surface architectural flaws across cloud-native services. Spearheaded SAST pipeline integration, detecting and remediating 150+ critical vulnerabilities early in the SDLC.
- Applied STRIDE threat modeling and MITRE ATT&CK frameworks to proactive threat hunts across cloud environments, enhancing service security posture by 10-15% through systematic vulnerability remediation and attack surface reduction.
- Engineered Python-based SIEM solutions and automated OS patching workflows into CI/CD pipelines, improving incident response efficiency by 30% and reducing manual intervention across deployment cycles.
- Authored service-level security documentation and drove cross-functional knowledge-sharing initiatives with product and engineering teams, setting security standards, mentoring team members on secure SDLC practices, and improving security posture consistency across multiple AWS service teams.

Application Security Intern | Varian Medical Systems, Palo Alto, USA

June 2020 - September 2020

- Conducted application access security reviews and built RBAC controls across SAP, HR-Workday, and ServiceNow enterprise applications, reducing unauthorized access incidents by 20% across a 10,000+ employee workforce and aligning application-layer controls with industry access management standards.
- Hardened authorization for SAP S/4HANA, refining PFCG roles and audit controls within the CRM application, tightening privilege boundaries, and reducing access-vulnerability surface by 15% through systematic RBAC review.
- Built a ServiceNow ticketing automation prototype that eliminated workflow redundancy and saved the team 6 hours weekly, prototyping secure-by-default patterns for enterprise application teams.

Security Analyst | SPRITZ Security & Privacy Group, Università di Padova, Italy

June 2018 - September 2018

- Designed and implemented an Intrusion Detection System (IDS) for Smart Grid architectures targeting Sybil and Wormhole attacks in RPL-based Industrial IoT networks, achieving 98% detection accuracy via the Highest Rank Common Ancestor verification principle and a proprietary wormhole detection algorithm integrated into the CONTIKI-OS COOJA simulator across modelled Wireless Sensor Network attack scenarios.

RESEARCH & OPEN SOURCE CONTRIBUTIONS

AI Agent Research & Development

Ongoing

- Building open-source AI agent toolkits and agentic harnesses at the intersection of multi-framework orchestration and security. Asobiba unifies factory abstractions across AutoGen, PydanticAI, SmolAgents, OpenAI Agents SDK, and LlamaIndex with a security tooling layer for AI-assisted offensive workflows.
- F.R.I.D.A.Y. and pi-friday extend this into agentic research systems with web intelligence, semantic caching, and multi-agent orchestration.

Financial Intelligence & Software Development

Ongoing

- Developing Kinyu, a financial data and AI research toolkit covering listed and unlisted market analysis, structuring fragmented company filings, audit reports, and management signals into actionable investment intelligence.
- Backed by Chishikiko, a software development knowledge base spanning Python, Golang, ML, and systems programming fundamentals that feeds directly into production tooling.

Security Research & CTF

Ongoing

- Conducting active offensive security research through Boxes and CTF, solving HackTheBox and TryHackMe challenges by combining traditional pentesting methodology (Nmap, Metasploit, Burp Suite, SQLMap) with companion AI tooling for automated enumeration and exploit development.
- Producing structured community writeups that document methodology and findings across completed engagements.

EDUCATION

M.S., Project Management - Cyber Security Operations, Harrisburg University

August 2024 - July 2025

Coursework: Agile/Scrum Methodology, Securing App and Software, Cloud and Network Security

GPA: 3.8/4.00

M.S., Cyber Security Engineering, University of Southern California (USC)

August 2019 - May 2021

Coursework: Web Application Security, Security Assurance & Privacy, Cryptography

GPA: 3.85/4.00

B.Tech., Electronics & Communication Engineering, Delhi Technological University

August 2015 - May 2019

Coursework: Computer Networks, Operating Systems, Algorithms & Data Structures

GPA: 8.22/10.00